B. Before an Exam

time limit per test: 0.5 second
memory limit per test: 64 megabytes

Tomorrow Peter has a Biology exam. He does not like this subject much, but d days ago he learnt that he would have to take this exam. Peter's strict parents made him prepare for the exam immediately, for this purpose he has to study not less than $minTime_i$ and not more than $maxTime_i$ hours per each i-th day. Moreover, they warned Peter that a day before the exam they would check how he has followed their instructions.

So, today is the day when Peter's parents ask him to show the timetable of his preparatory studies. But the boy has counted only the sum of hours $sumTime$ spent him on preparation, and now he wants to know if he can show his parents a timetable $schedule$ with d numbers, where each number $schedule_i$ stands for the time in hours spent by Peter each i-th day on biology studies, and satisfying the limitations imposed by his parents, and at the same time the sum total of all $schedule_i$ should equal to $sumTime$.

Input

The first input line contains two integer numbers $d, sumTime$ $(1 \le d \le 30, 0 \le sumTime \le 240)$ — the amount of days, during which Peter studied, and the total amount of hours, spent on preparation. Each of the following d lines contains two integer numbers $minTime_i, maxTime_i$ $(0 \le minTime_i \le maxTime_i \le 8)$, separated by a space — minimum and maximum amount of hours that Peter could spent in the i-th day.

Output

In the first line print `YES`, and in the second line print d numbers (separated by a space), each of the numbers — amount of hours, spent by Peter on preparation in the corresponding day, if he followed his parents' instructions; or print `NO` in the unique line. If there are many solutions, print any of them.

Examples

input Copy

```
1 48
5 7
```

output Copy

```
NO
```

input Copy

```
2 5
0 1
3 5
```

output Copy

```
YES
1 4
```

Codeforces Beta Round 4 (Div. 2 Only)
Finished
Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
368350517	Mar/27/2026 10:40	Accepted

→ Problem tags

constructive algorithms greedy *1200

No tag edit access